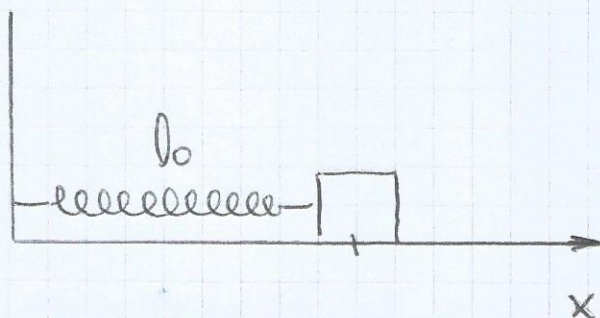
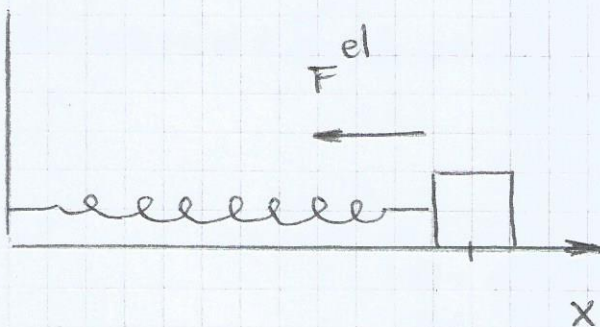
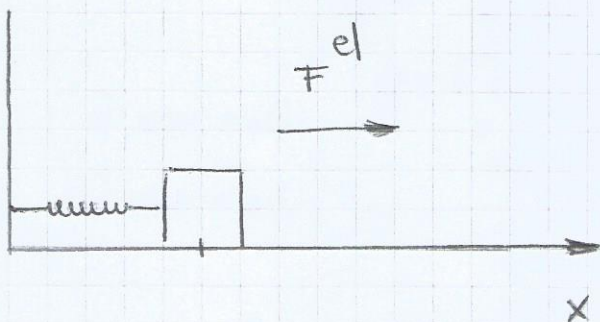


Unidad IV: Movimiento oscilatorio. l_0 , longitud natural $x > l_0$  $x < l_0$

$$\overline{F}^{el} = -K \underbrace{(x - l_0)}_{\text{elongación}} \hat{x} \quad \text{con } K, \text{ constante elástica.}$$

↳ cuanto se aparta de la posición de equilibrio.

$$\begin{aligned} \overline{F}^{el} &= m a \\ -K(x - l_0) \hat{x} &= m \ddot{x} \hat{x} \\ \ddot{x} + \frac{K}{m} x &= \frac{K}{m} l_0 \\ \ddot{x} + \omega_0^2 x &= \omega_0^2 l_0 \end{aligned}$$

$$\omega_0 = \sqrt{\frac{K}{m}} \quad (\text{pulsación})$$

Apéndice matemático:

Ecuación de Euler: $\ddot{x}(t) = \ddot{x} + b\dot{x} + cx$ con b y c constantes

Si $f(t) = 0$, es homogéneo
 $f(t) \neq 0$, es inhomogéneo.

La solución de la ecuación es:

$$x(t) = \underbrace{x_h(t)}_{\text{Homogéneo}} + \underbrace{x_p(t)}_{\text{Particular}}$$

$x_h(t)$, sc de $\ddot{x} + b\dot{x} + cx = 0$
 $x_p(t)$, sc de $\ddot{x} + b\dot{x} + cx = f(t)$

$$\begin{aligned}x_h &= e^{\lambda t} \\ \dot{x}_h &= \lambda e^{\lambda t} \\ \ddot{x}_h &= \lambda^2 e^{\lambda t}\end{aligned}$$

$$\begin{aligned}\ddot{x}_h + \omega_0^2 x_h &= 0 \\ \lambda^2 e^{\lambda t} + \omega_0^2 e^{\lambda t} &= 0 \\ e^{\lambda t} (\lambda^2 + \omega_0^2) &= 0 \\ \lambda^2 &= -\omega_0^2\end{aligned}$$

$$\lambda_1 = \omega_0 i$$

$$\lambda_2 = -\omega_0 i$$

$$x_h = A e^{i\omega_0 t} + B e^{-i\omega_0 t}$$

Ecuación de Moivre: $e^{ix} = \cos x + i \sin x$

$$x_h \begin{cases} A_1 \cos(\omega_0 t + \varphi) \\ A_2 \sin(\omega_0 t + \varphi) \end{cases}$$

$$x_p = c$$

$$\dot{x}_p = 0$$

$$\ddot{x}_p = 0$$

$$\ddot{x} + \omega_0^2 x = \omega_0^2 l_0$$

$$0 + \cancel{\omega_0^2} c = \cancel{\omega_0^2} l_0$$

$$c = l_0$$

$$x(t) = A \cos(\omega_0 t + \varphi) + l_0$$

$$\dot{x}(t) = -A\omega_0 \sin(\omega_0 t + \varphi)$$

Condiciones iniciales:

$$x(t=0) = x_0 \quad x_0 = A \cos \varphi + l_0 \quad (1)$$

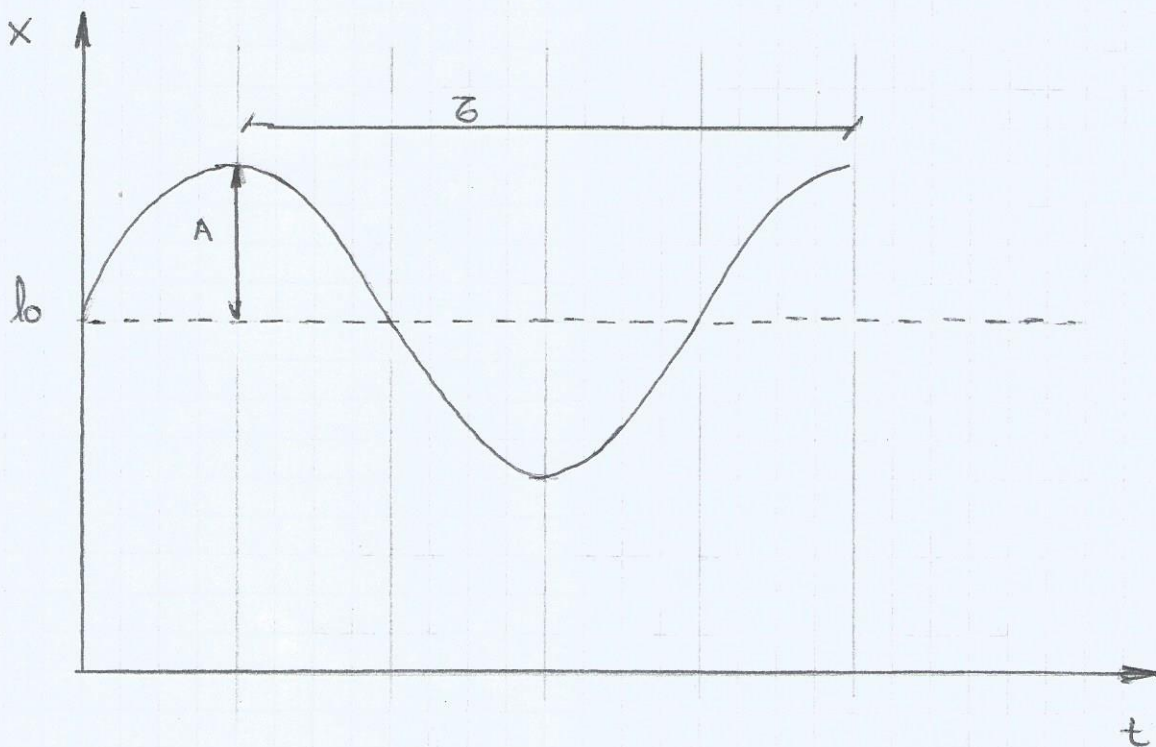
$$v(t=0) = v_0 \quad v_0 = -A\omega_0 \sin \varphi \quad (2)$$

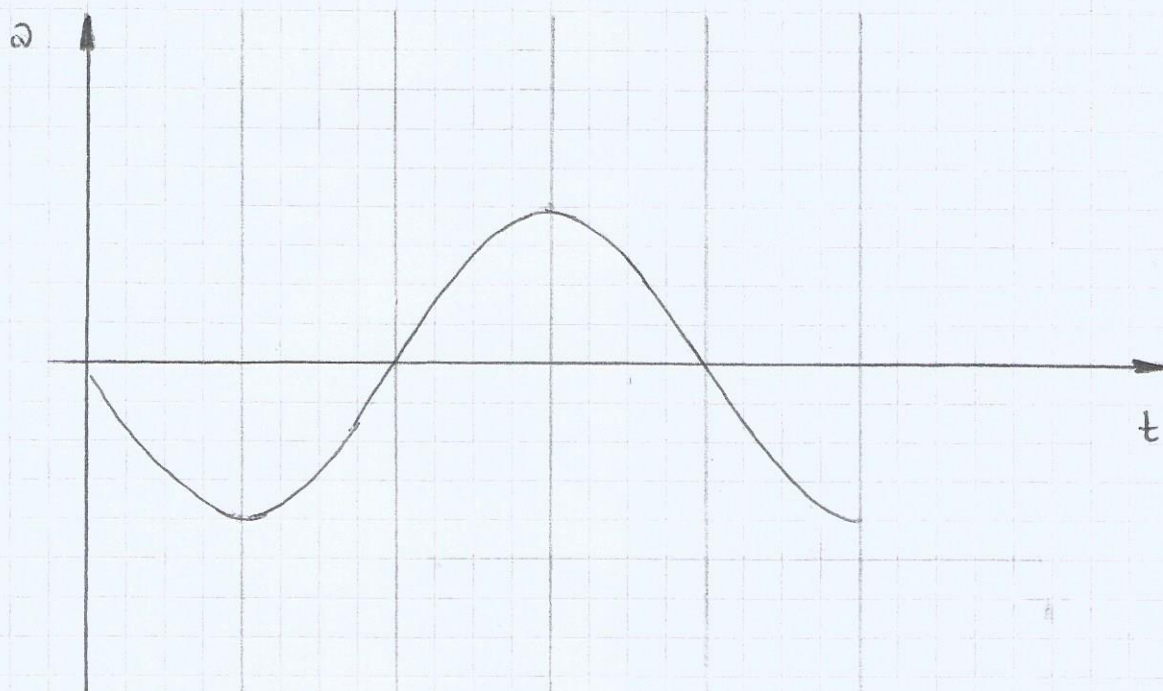
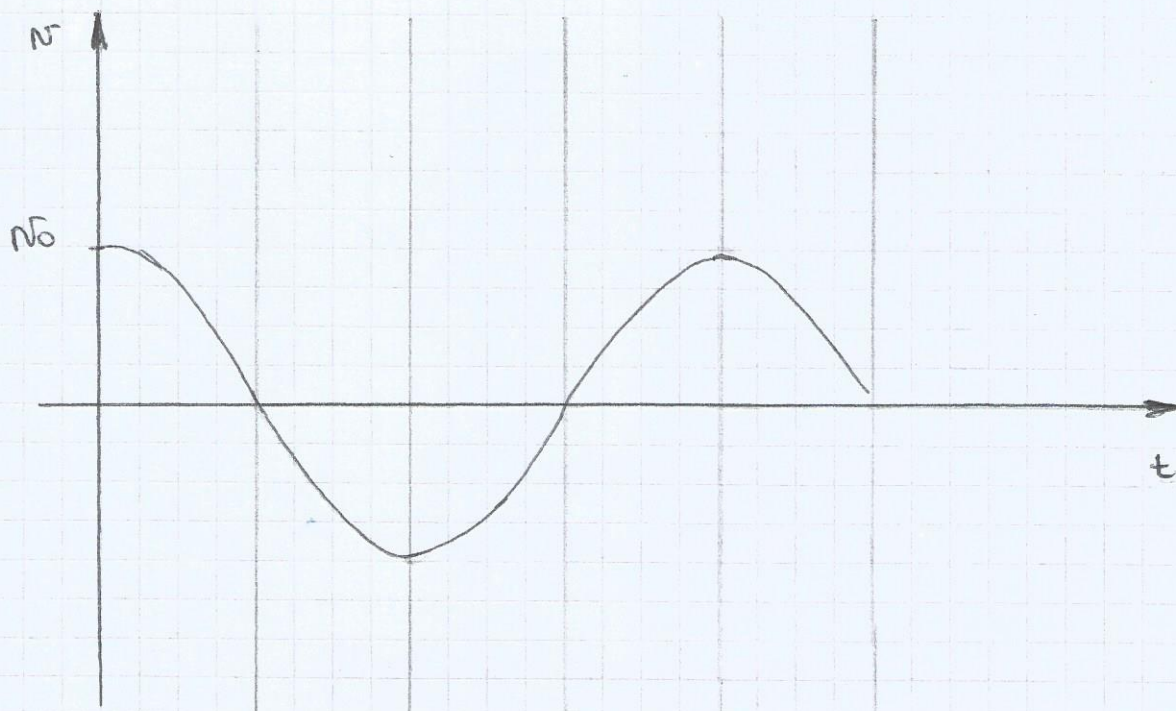
$$(2)/(1) \quad \frac{v_0}{x_0 - l_0} = -\frac{A\omega_0 \sin \varphi}{A \cos \varphi} = -\omega_0 \tan \varphi$$

$$\varphi = \arctan\left(\frac{-v_0}{\omega_0(x_0 - l_0)}\right)$$

$$(1) \quad x_0 = A \cos \varphi + l_0$$

$$A = \frac{x_0 - l_0}{\cos \varphi}$$





$$N=0$$

$$Q = \text{máx}$$

$$-A$$

$$N = \text{máx}$$

$$Q=0$$

$$l_0$$

$$N=0$$

$$Q = \text{máx}$$

$$A$$

$$x(t) = A \cos(\omega_0 t + \varphi) + l_0$$

$$x(t+\tau) = A \cos(\omega_0(t+\tau) + \varphi) + l_0$$

$$\omega_0 t + \varphi + 2\pi = \omega_0 t + \omega_0 \tau + \varphi$$

$$2\pi = \omega_0 \tau$$

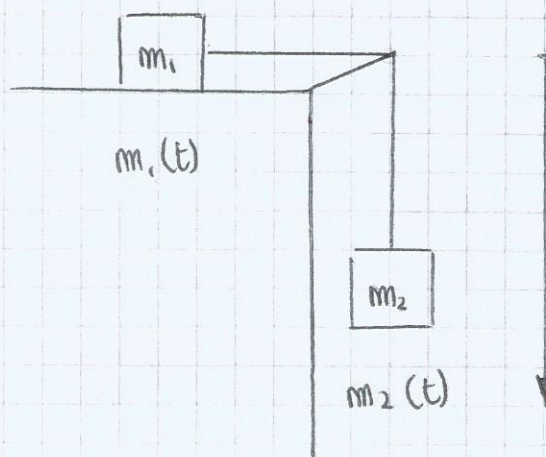
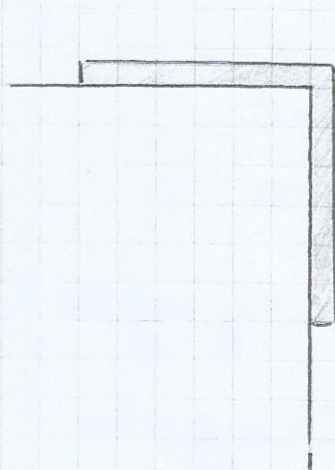
$$\tau = \frac{2\pi}{\omega_0}$$

Soluciones de:

$$\ddot{x} - \omega_0^2 x = 0$$

$$x_H(t) = A_1 \cosh(\omega_0 t + \varphi)$$

$$x_H(t) = A_2 \sinh(\omega_0 t + \varphi)$$



densidad lineal de masa : $\lambda = \frac{m}{l} = \frac{mT}{LT}$

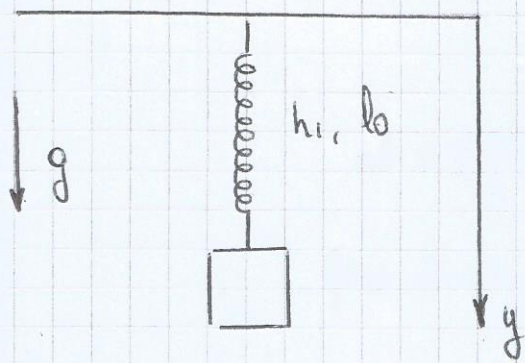
$$T = m_1 \ddot{x} = \lambda(L-x) \ddot{x} \quad (1)$$

$$P - T = m_2 \ddot{x} = \lambda x \ddot{x} \quad (2)$$

$$(1) + (2) \quad \lambda x g = (\lambda(L-x) + \lambda x) \ddot{x}$$

$$\lambda x g = \lambda L \ddot{x}$$

$$\ddot{x} - \frac{g}{L} x = 0 \Rightarrow \text{forma de un oscilador}$$



$$mg - k(y - l_0) = m \ddot{y}$$

$$y - \frac{k}{m}(y - l_0) = \ddot{y}$$

$$\ddot{y} + \frac{k}{m}y = g + \frac{k}{m}l_0$$

Busco las soluciones:

$$y_H = A \cos(\omega_0 t + \varphi)$$

$$y_P = \frac{mg}{k} + l_0$$

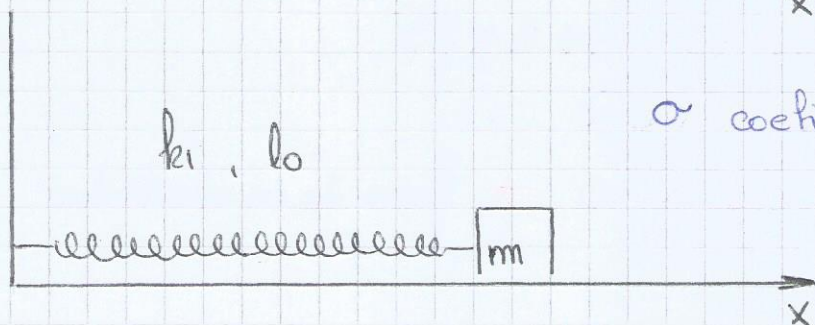
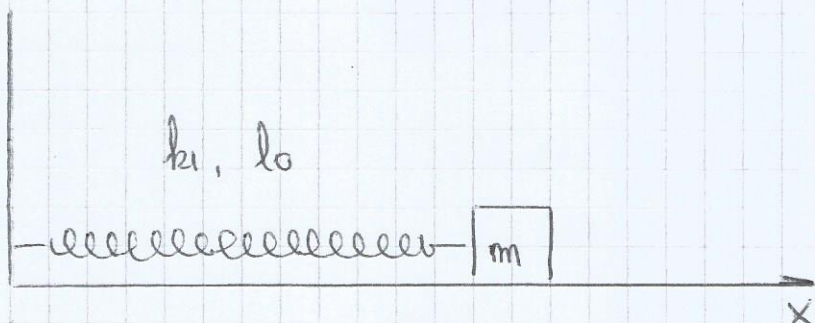
$$y(t) = A \cos(\omega_0 t + \varphi) + \frac{mg}{k} + l_0$$

$$y_P = C$$

$$\ddot{y}_P = 0$$

$$\frac{k}{m}C = g + \frac{k}{m}l_0$$

$$C = \frac{mg}{k} + l_0$$



α coeficiente de viscosidad

$$F^{\text{el}} = -kx \hat{x}$$

$$F_v = -\sigma v \hat{x}$$

$$-kx - \sigma \dot{x} = m \ddot{x}$$

$$\ddot{x} + \frac{\sigma}{m} \dot{x} + \frac{k}{m} x = 0$$

$$x(t) = x_H$$

Suponho $\omega_0^2 = \frac{k}{m}$; $2b = \frac{\sigma}{m}$

$$\ddot{x} + 2b \dot{x} + \omega_0^2 x = 0$$

$$\lambda^2 e^{\lambda t} + 2b \lambda e^{\lambda t} + \omega_0^2 e^{\lambda t} = 0$$

$$e^{\lambda t} (\lambda^2 + 2b\lambda + \omega_0^2) = 0$$

$$\lambda = \frac{-2b \pm \sqrt{(2b)^2 - 4 \cdot 1 \cdot \omega_0^2}}{2}$$

$$\lambda = -b \pm \sqrt{b^2 - \omega_0^2}$$

$$x \sim e^{\lambda t}$$

$$\dot{x} \sim \lambda e^{\lambda t}$$

$$\ddot{x} \sim \lambda^2 e^{\lambda t}$$

i) Caso I: $\lambda_1 \neq \lambda_2$, $\lambda_{1,2} \in \mathbb{R} \Rightarrow b > \omega_0$

ii) Caso II: $\lambda_1 \neq \lambda_2$, $\lambda_{1,2} \in \mathbb{C} \Rightarrow b < \omega_0$

iii) Caso III: $\lambda_1 = \lambda_2 = -b$, $b = \omega_0$

Caso I: $b > \omega_0 \Rightarrow \frac{\sigma}{2m} > \sqrt{\frac{k}{m}}$

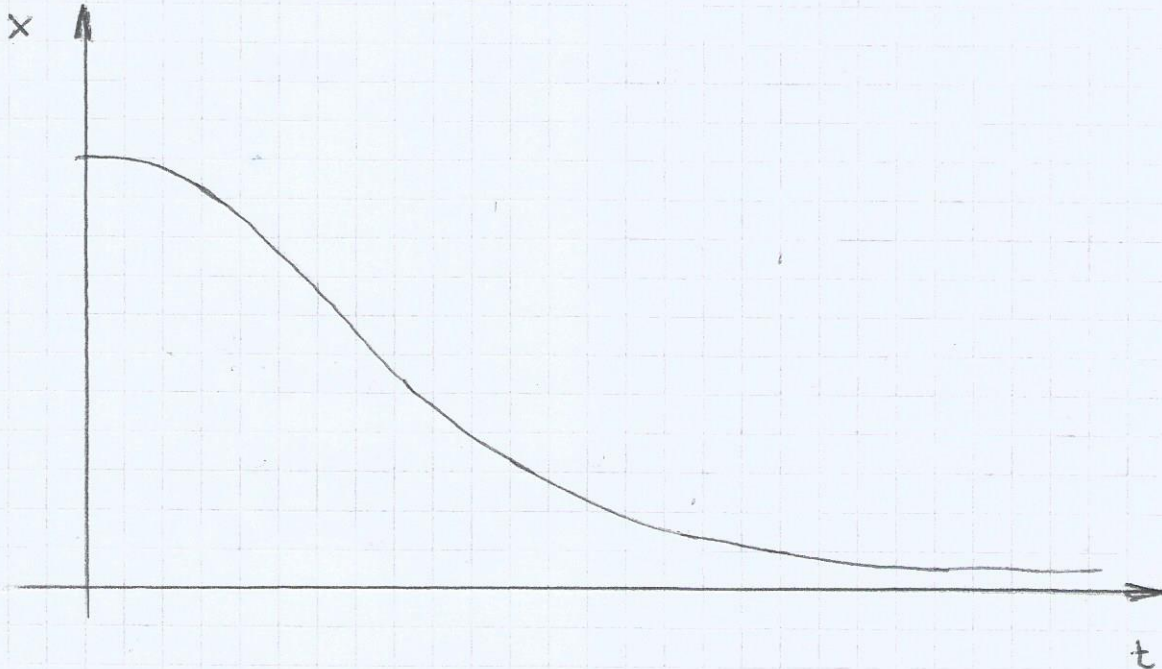
$$x(t) = A e^{\lambda_1 t} + B e^{\lambda_2 t}$$

$$x(t) = A e^{-bt + \sqrt{b^2 - \omega_0^2} t} + B e^{-bt - \sqrt{b^2 - \omega_0^2} t}$$

$$x(t) = e^{-bt} (A e^{\sqrt{b^2 - \omega_0^2} t} + B e^{-\sqrt{b^2 - \omega_0^2} t})$$

$$x(t) \begin{cases} e^{-bt} (C \cosh(\sqrt{b^2 - \omega_0^2} t + \varphi)) \\ e^{-bt} (D \sinh(\sqrt{b^2 - \omega_0^2} t + \varphi)) \end{cases}$$

Movimiento sobreamortiguado.

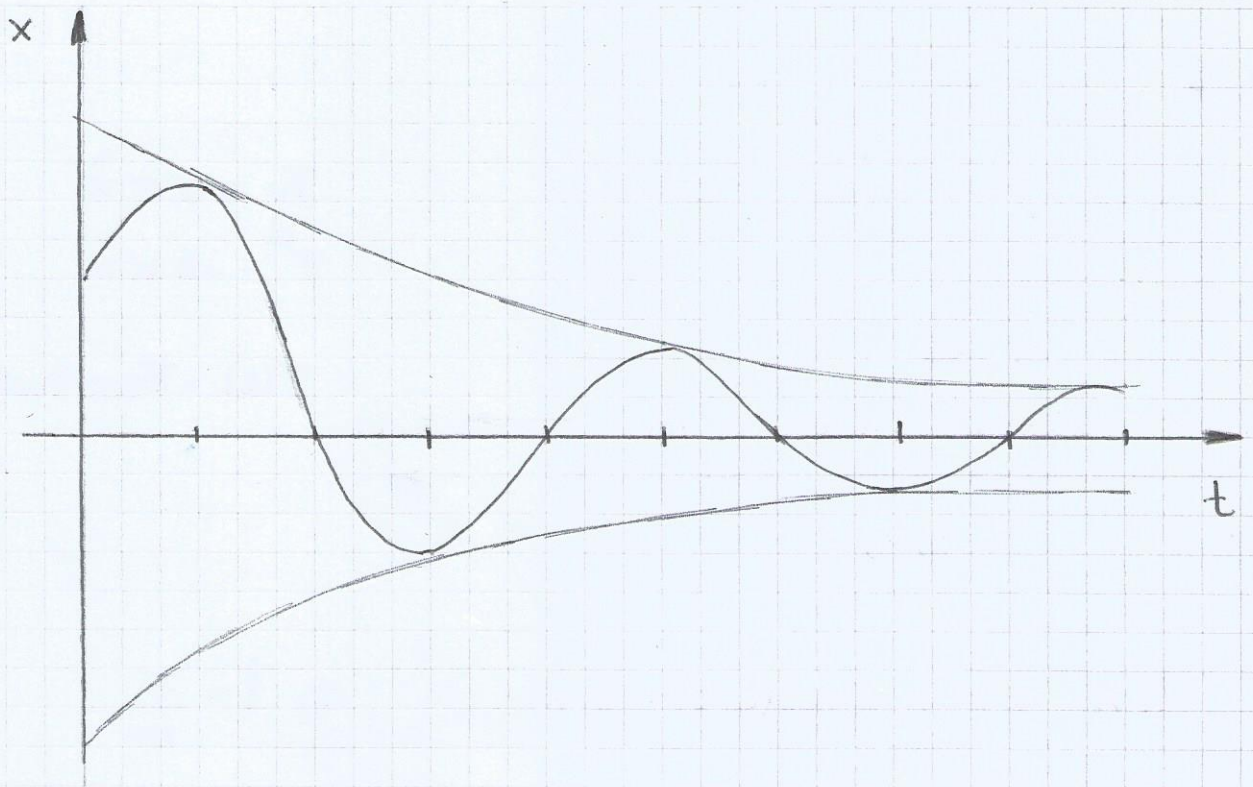


Caso II: $b < \omega_0 \Rightarrow \frac{\gamma}{m} < \sqrt{\frac{k}{m}}$

$$x(t) = A e^{-bt + i \sqrt{\omega_0^2 - b^2} t} + B e^{-bt - i \sqrt{\omega_0^2 - b^2} t}$$

$$x(t) = e^{-bt} (A e^{i \sqrt{\omega_0^2 - b^2} t} + B e^{-i \sqrt{\omega_0^2 - b^2} t})$$

$$x(t) \begin{cases} C e^{-bt} \cos(\sqrt{\omega_0^2 - b^2} t + \varphi) \\ D e^{-bt} \sin(\sqrt{\omega_0^2 - b^2} t + \varphi) \end{cases}$$



Movimiento amortiguado:

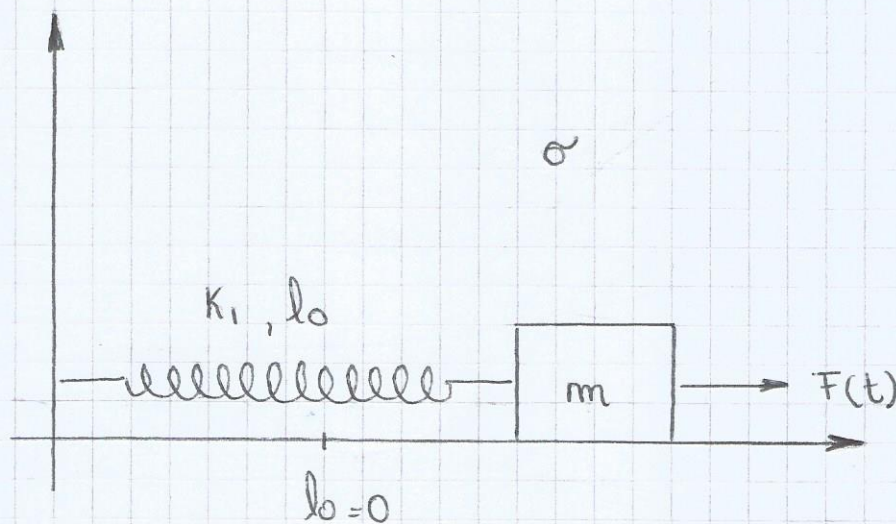
$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}} = \frac{2\pi}{\omega_0 \cdot \sqrt{1 - \frac{b^2}{\omega_0^2}}}$$

Caso III: $b = \omega_0 \Rightarrow \frac{\sigma}{2m} \approx \sqrt{\frac{k}{m}}$

$$x(t) = e^{-bt} (A + Bt)$$



Oscilaciones forzadas:



$$\vec{F}_v = -\sigma \vec{v}$$

$$\vec{F}^{el} = -k \vec{x}$$

$$F(t) = F \cos(\omega t)$$

$$-Kx - \sigma \dot{x} + F \cos(\omega t) = m \ddot{x}$$

$$\text{Si } f = \frac{F}{m}$$

$$\ddot{x} + \frac{\sigma}{m} \dot{x} + \frac{k}{m} x = f \cos(\omega t)$$

$$x(t) = x(t)_h + x(t)_p$$

i) Transitoria: $x(t) = x_h + x_p$

ii) Estacionaria: $x(t) = x_p$

$$x_p = A \cos(\omega t) + B \sin(\omega t)$$

$$\dot{x}_p = -A \omega \sin(\omega t) + B \omega \cos(\omega t)$$

$$\ddot{x}_p = -A \omega^2 \cos(\omega t) - B \omega^2 \sin(\omega t)$$

Reemplazo en ec.

$$\ddot{x} + \frac{\sigma}{m} \dot{x} + \frac{k}{m} x = f \cos(\omega t)$$

$$[(\omega_0^2 - \omega^2)A + 2b\omega B] \cos(\omega t) + [(\omega_0^2 - \omega^2)B - 2b\omega A] \sin(\omega t) = f \cos(\omega t)$$

$$\text{Con } t=0 \Rightarrow (\omega_0^2 - \omega^2)A + 2b\omega B = f$$

$$\text{Con } t = \frac{\pi}{2\omega} \Rightarrow (\omega_0^2 - \omega^2)B - 2b\omega A = 0$$

$$A = \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2} f \quad B = \frac{2b\omega}{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2} f$$

$$x_p = \frac{f}{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2} [(\omega_0^2 - \omega^2) \cos \omega t + 2b\omega \sin \omega t]$$

Si encuentro φ tal que:

$$\cos \varphi = \frac{(\omega_0^2 - \omega^2)}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2}} \quad \sin \varphi = \frac{2b\omega}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2}}$$

Podemos reescribir

$$[\cos \varphi \cdot \cos \omega t + \sin \varphi \sin \omega t] \cdot \frac{f}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2}}$$

$$\frac{f}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2b\omega)^2}} \cdot \cos(\omega t - \varphi) \quad \varphi = \arctg \frac{2b\omega}{\omega_0^2 - \omega^2}$$

i) $\omega \ll \omega_0$

$$\left. \begin{array}{l} \sin \varphi \rightarrow 0 \\ \cos \varphi \rightarrow 1 \end{array} \right\} \varphi \approx 0$$

$$A \rightarrow \frac{f}{\omega_0^2} = \frac{F}{k}$$

ii) $\omega \gg \omega_0$

$$\left. \begin{array}{l} \sin \varphi \rightarrow 0 \\ \cos \varphi \rightarrow -1 \end{array} \right\} \varphi \approx \pi$$

$$A \rightarrow \frac{f}{\omega^2} = \frac{F}{m \omega^2}$$

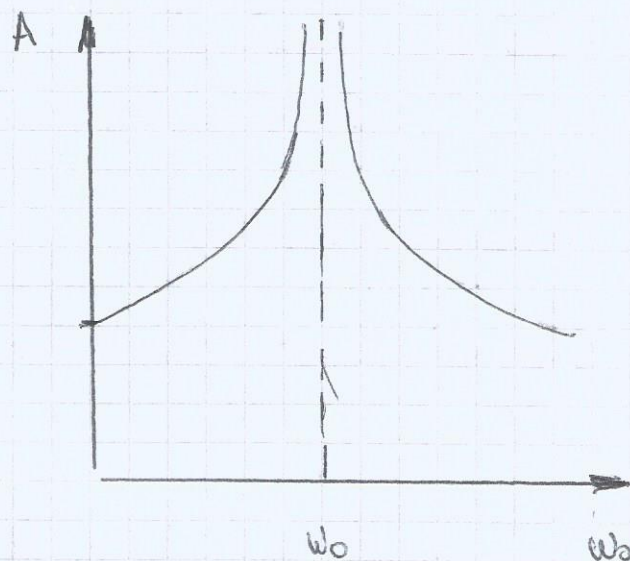
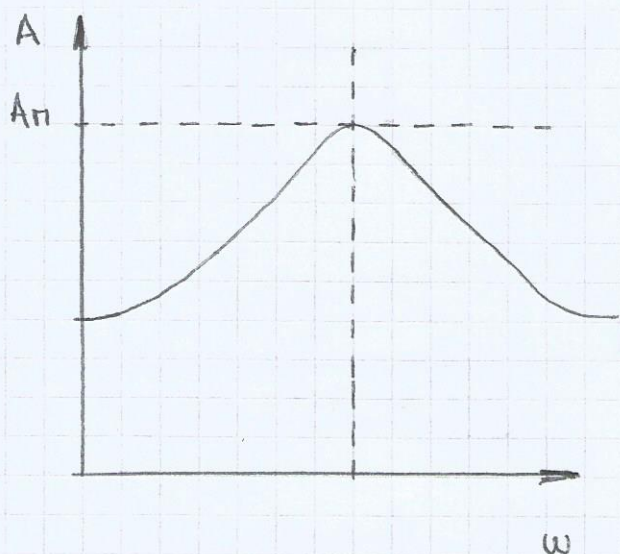
F para el mínimo A

$$A = \frac{f}{[(\omega_0^2 - \omega^2)^2 + (2b\omega)^2]^{\frac{1}{2}}}$$

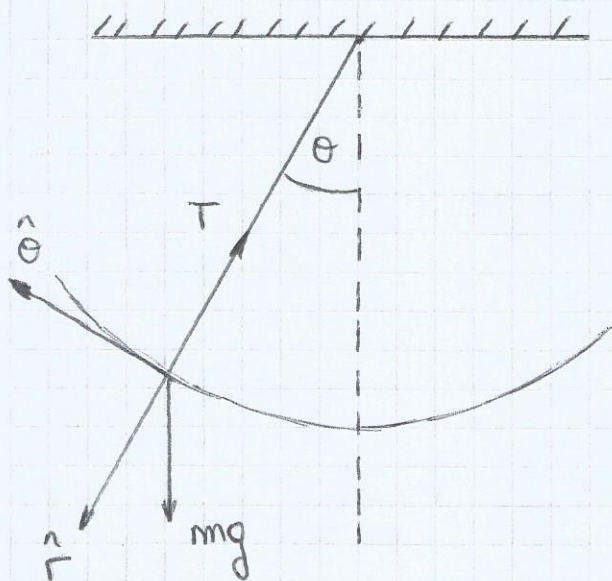
$$\frac{d}{d\omega} ((\omega_0^2 - \omega^2)^2 + (2b\omega)^2) = 0$$

$$\omega^2 = \omega_0^2 - 2b$$

$$A_m = \frac{f}{2b[\omega_0^2 - b^2]^{\frac{1}{2}}}$$



Péndulo físico:



$$\hat{r}: mg \cos \theta - T = -ml \ddot{\theta}^2$$

$$\hat{\theta}: -mg \sin \theta = ml \ddot{\theta}$$

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

Si desarrollamos por Taylor el $\sin \theta$ con $\theta \approx 0$

$$\sin \theta \approx \theta - \frac{1}{3!} \theta^3 + \dots + \dots \Rightarrow \sin \theta = \theta$$

$$\ddot{\theta} + \frac{g}{l} \cdot \theta = 0 \quad \omega_0 = \sqrt{\frac{g}{l}} \quad T = \frac{2\pi}{\omega_0} = 2\pi \sqrt{\frac{l}{g}}$$

Solución general:

$$\theta(t) = A \cos \left(\sqrt{\frac{g}{l}} t + \varphi \right)$$